AMERICAN INTERNATIONAL UNIVESITY-BANGLADESH

Metro Rail Management System

Name	ID	Contribution
Basudeb Kundu	23-50856-1	ER Diagram, Normalization, Schema Diagram,Table Creation, Data Insertion, QueryWriting,Relational Algebra
Ekramul Hasib	23-51106-1	Introduction, Scenario Description, Schema Diagram, Relational Algebra
Majumder Partha Sarothi	22-48190-2	Query Writing, Relational Algebra
Md Raihaan Ahmed Raafi	23-50489-1	Query Writing, Relational Algebra

Course Title : INTRODUCTION TO DATABASE.
Section : [J]

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Introduction:

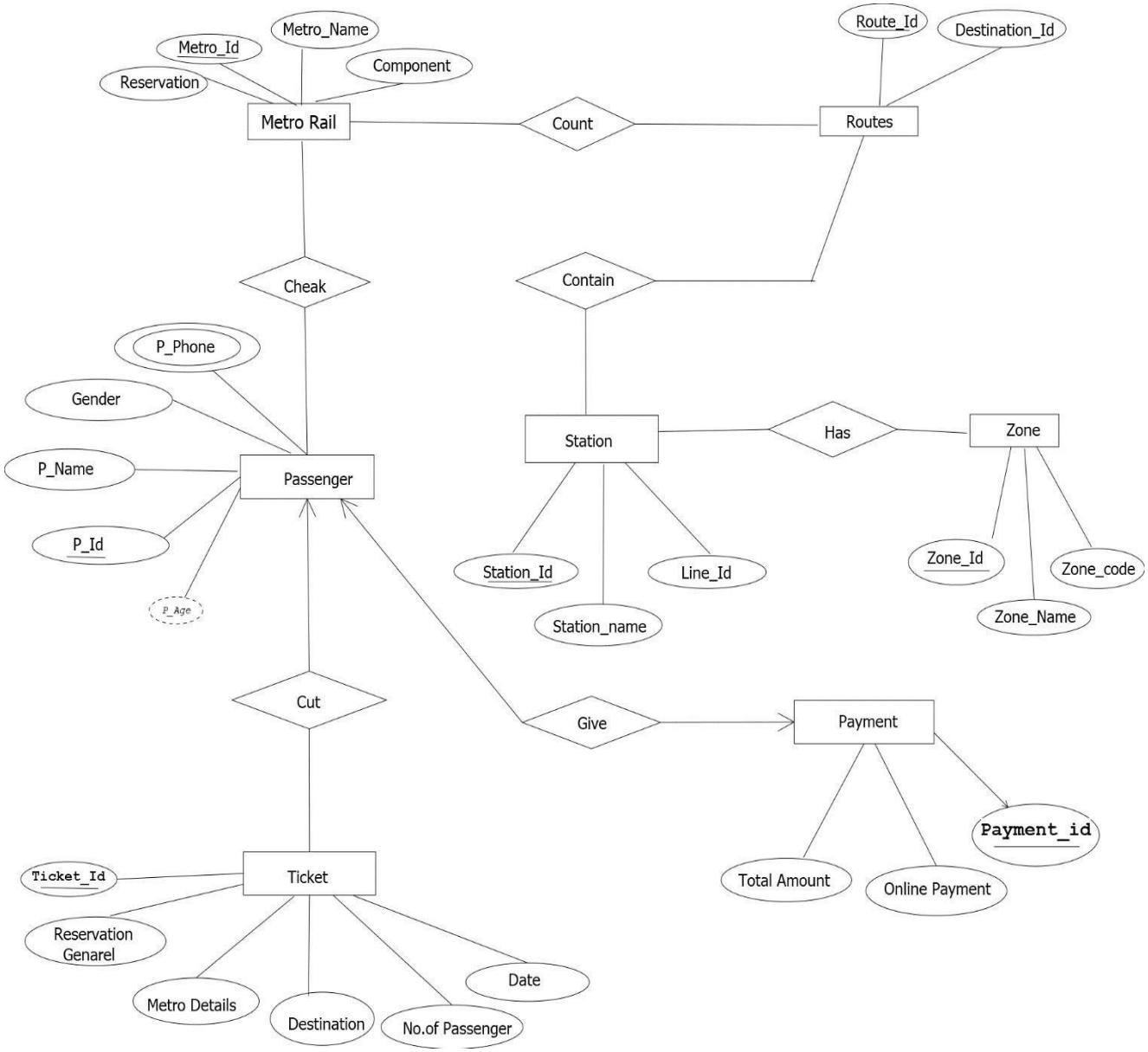
The Metro Rail Database Project is a comprehensive system developed to streamline the management and operation of a metro rail network. It serves as a centralized repository for storing, organizing, and accessing crucial data related to various aspects of the metro system. This project encompasses the creation of a robust database architecture capable of handling information such as train schedules, routes, fares, station details, passenger statistics, maintenance records, and real-time updates. With a user-friendly interface, the project enables passengers to effortlessly access essential information like schedules, routes, and fares. Moreover, it offers administrative tools empowering system administrators to efficiently manage and update the database, ensuring accurate and timely information dissemination. Key features of the Metro Rail Database Project include real-time integration for updates on train schedules and service disruptions, passenger services like trip planning and ticket purchasing, and analytical tools for optimizing system performance and passenger experience. Additionally, stringent security measures are implemented to safeguard sensitive data and maintain the integrity of the system.

Scenario Description

StudentID1: 23-50856-1 Name: Basudeb Kundu	StudentID:23-51106-1 Name Ekramul Hasib
StudentID:22-48190-2 Name: Partha Sharothi Mazumder	StudentID4:23-50489-1 Name: Md Raihaan Ahmed Raafi
CO2: Understand the fundamental concepts underlying database systems and gain hands-on experience with ER diagram Case study	
PO-c2: Develop process for complex computer science and engineering problems considering cultural and societal factors.	Marks

In a metro rail management system, where passengers can make reservations for tickets. Each passenger is identified by a unique Passenger Id, name, phone number, and age. The metro system has multiple lines, each with a unique Line Id, and each line has multiple routes, identified by a Route Id. Each route contains multiple stations, each with a unique Station_Id and name. Passengers can make reservations for a specific component on a specific metro line, and choose their source and destination stations. The reservation is associated with a specific date and zone, which determines the fare for the trip. Passengers can pay for their tickets using online payment, and the total amount due is calculated by the payment system. Passengers can check the status of their reservation using the Cheak function, and obtain their ticket at the station by providing their reservation details.

Entity Relationship Diagram:



ER Diagram For Metro Rail Management System

Normalization:

UNF

Count (Metro_Id,Metro_Name,Reservation,Component,Route_Id,Destination_Id)

1NF

There is no multi valued attribute. Relation already in 1NF.

1. Metro_Id, Metro_Name,Reservation,Component,Route_Id,Destination_Id

2NF

1.Metro_Id, Metro_Name, Reservation, Component

2.Route_Id, Destination_Id

3NF

There is no transitive dependency. Relation already in 3NF.

1.Metro_Id, Metro_Name, Reservation, Component

2.Route_Id, Destination_Id

Table Creation

1.Metro_Id, Metro_Name, Reservation, Component

2.Route_Id, Destination_Id,**Metro_Id**.

UNF

Cheak (Metro_Id, Metro_Name, Reservation, Component, P_Id, P_Age,P_Name, Gender, P_phone)

1NF

P_phone is a multi valued attribute.

1. Metro_Id, Metro_Name, Reservation, Component, P_Id, P_Age, P_Name, Gender, P_phone.

2NF

1. Metro_Id, Metro_Name, Reservation, Component.

2. P_Id, P_Age, P_Name, Gender, P_phone.

3NF

There is no transitive dependency. Relation already in 3NF.

1. Metro_Id, Metro_Name, Reservation, Component.

2. P_Id,P_Age,P_Name,Gender,P_phone.

Table Creation

1. Metro_Id, Metro_Name, Reservation, Component.

2. P_Id, P_Age, P_Name, Gender, P_phone. **Metro_Id**

UNF

Cut (P_Id, P_Age, P_Name, Gender, P_phone, Ticket_Id, Reservation Genarel ,Metro Details, Destination, No.Of Passenger, Travel_Date)

1NF

P_phone is a multi valued attribute.

1. P_Id , P_Age, P_Name,Gender, P_phone, Ticket_Id , Reservation Genarel, Metro Details, Destination,No.Of Passenger, Travel_Date

2NF

1.P_Id, P_Age, P_Name, Gender, P_phone, Reservation

2. Ticket_Id, Reservation Genarel, Metro Details, Destination, No.Of Passenger, Travel_Date

3NF

There is no transitive dependency. Relation already in 3NF.

1.P_Id, P_Age, P_Name, Gender, P_phone, Reservation

2. Ticket_Id, Reservation Genarel, Metro Details, Destination, No.Of Passenger, Travel_Date

Table Creation

1.P_Id, P_Age, P_Nam, Gender, P_phone

2. Ticket_Id, Reservation Genarel, Metro Details, Destination, No.Of Passenger, Travel_Date, **P_Id.**

UNF

Give (P_Id, P_Age, P_Name, Gender, P_phone, Payment_Id ,Total Amount,Online

Payment)

1NF

P_phone is a multi valued attribute.

1. P_Id, P_Age, P_Name, Gender, P_phone, Payment_Id, Total Amount, Online Payment

2NF

1. P_Id, P_Age, P_Name, Gender, P_phone,

2. Payment_Id ,Total Amount, Online Payment.

3NF

There is no transitive dependency. Relation already in 3NF.

1. P_Id, P_Age, P_Name, Gender, P_phone,

2. Payment_Id ,Total Amount,Online Payment.

Table Creation

1. P_Id, P_Age, P_Name, Gender, P_phone,

2. Payment_Id,Total Amount, Online Payment, **P_Id**

UNF

Has (Station_Id, Station_Name, Line_ID, Zone_Id, Zone_Name, Zone_Code)

1NF

There is no multi valued attribute. Relation already in 1NF.

1**.** Station_Id, Station_Name, Line_ID, Zone_Id, Zone_Name, Zone_Code

2NF

1. Station_Id, Station_Name, Line_id

2. Zone_Id, Zone_Name, Zone_Code

3NF

There is no transitive dependency. Relation already in 3NF.

1. Station_Id, Station_Name, Line_id

2. Zone_Id, Zone_Nam, Zone_Code, **Station_Id**

Table Creation

1. Station_Id, Station_Name, Line_Id

2. Zone_Id, Zone_Name, Zone_Code, **Station_Id**

UNF

Contain (Route_Id, Destination_Id, Station_Id, Station_Name, Line_Id)

1NF

There is no multi valued attribute. Relation already in 1NF.

1.Route_Id, Destination_Id, Station_Id, Station_Name, Line_Id

2NF

1. Route_Id, Destination_Id,

2. Station_Id, Station_Name, Line_Id

3NF

There is no transitive dependency. Relation already in 3NF.

1. Route_Id, Destination_Id,

2. Station_Id, Station_Name, Line_Id

Table Creation

1. Route_Id, Destination_Id

2. Station_Id, Station_Name, Line_Id. **Route_Id,**

Temporary Tables

1.Metro_Id, Metro_Name, Reservation, Component

2.Route_Id, Destination_Id, **Metro_Id**.

3. ~~Metro_Id,Metro_Name,Reservation,Component.~~

4.P_Id,P_Age,P_Name,Gender,P_phone. **Metro_Id**

5.~~P_Id, P_Age,P_Name,Gender,P_phone~~

6.Ticket_Id, Reservation Genarel, Metro Details, Destination, No.Of Passenger, Travel_Date,**P_Id.**

7.~~P_Id, P_Age, P_Name, Gender, P_phone,~~

8.Payment_Id,Total Amount,Online Payment, **P_Id**

9.~~Station_Id, Station_Name. Line_Id~~

10. Zone_Id, Zone_Name, Zone_Code, **Station_Id**

11.~~Route_Id, Destination_Id,~~

12.Station_Id, Station_Name, Line_Id. **Route_Id,**

Final Tables

1. Metro_Id, Metro_Name, Reservation, Component
2. Route_Id, Destination_Id, **Metro_Id**.
3. P_Id, P_Name P_Age, Gender, P_phone. **Metro_Id**
4. Ticket_Id, Travel_Date, Num_Passenger, Destination, Metro Details, Resurvation_Status, **P_Id.**
5. Payment_Id, Online Payment, Total Amount, **P_Id**
6. Zone_Id, Zone_Name, Zone_Code, **Station_Id**
7. Station_Id, Station_Name, Line_Id. **Route_Id,**

Schema Diagram :

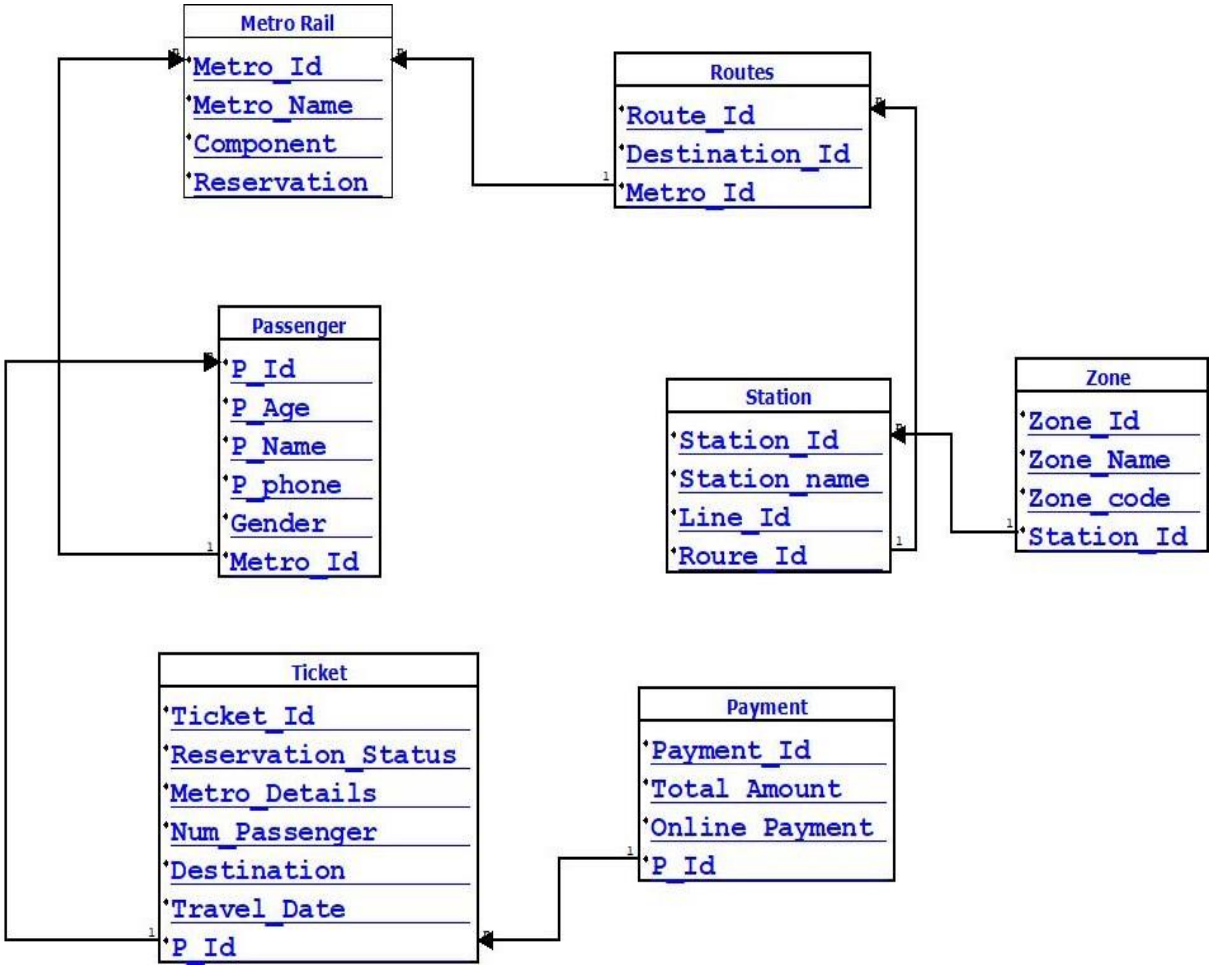


Table Creation

Student Id1:23- 50856-1 Name: Basudeb kudnu	StudentID3: 22-48190-2 Name: Partha Sharothi Mazumder
StudentID2: 23-51106-1 Name: Ekramul Hasib	StudentID4: 23-50489-1 Name: Md Raihaan Ahmed Raafi
CO4: Creating DML, DDL using Oracle and connection with ODBC/JDBC for existing JAVA application	
PO-e-2: Use modern engineering and IT tools for prediction and modeling of complex computer science and engineering problem	Marks

Table Creation:

Create User:

```
CREATE USER metro_rail_ms IDENTIFIED BY basu;
GRANT CONNECT, RESOURCE, UNLIMITED TABLESPACE TO metro_rail_ms;
GRANT CREATE VIEW TO metro_rail_ms;
```

Metro rail table :

Autocommit Display 10

```
CREATE TABLE metro_rail (
    metro_id NUMBER(10) PRIMARY KEY,
    metro_name VARCHAR2(50),
    reservation VARCHAR2(20),
    component VARCHAR2(50)
);

desc metro_rail;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **METRO_RAIL**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
METRO_RAIL	METRO_ID	Number	-	10	0	1	-	-	-
	METRO_NAME	Varchar2	50	-	-	-	✓	-	-
	RESERVATION	Varchar2	20	-	-	-	✓	-	-
	COMPONENT	Varchar2	50	-	-	-	✓	-	-

1 - 4

Routes Table:

Autocommit Display 10

```
CREATE TABLE routes (
    route_id NUMBER(5) PRIMARY KEY,
    destination_id NUMBER(5),
    metro_id NUMBER(10),
    FOREIGN KEY (metro_id) REFERENCES metro_rail(metro_id)
);

desc routes;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **ROUTES**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
ROUTES	ROUTE_ID	Number	-	5	0	1	-	-	-
	DESTINATION_ID	Number	-	5	0	-	✓	-	-
	METRO_ID	Number	-	10	0	-	✓	-	-

1 - 3

Passenger Table:

```
CREATE TABLE passenger (
    p_id NUMBER(10) PRIMARY KEY,
    p_name VARCHAR2(100),
    p_age NUMBER(4),
    gender VARCHAR2(10),
    p_phone VARCHAR2(20),
    metro_id NUMBER(10),
    FOREIGN KEY (metro_id) REFERENCES metro_rail(metro_id)
);
desc passenger;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **PASSENGER**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PASSENGER	P_ID	Number	-	10	0	1	-	-	-
	P_NAME	Varchar2	100	-	-	-	✓	-	-
	P_AGE	Number	-	4	0	-	✓	-	-
	GENDER	Varchar2	10	-	-	-	✓	-	-
	P_PHONE	Varchar2	20	-	-	-	✓	-	-
	METRO_ID	Number	-	10	0	-	✓	-	-

1 - 6

Ticket Table:

Autocommit Display 10

```
CREATE TABLE ticket (
    ticket_id NUMBER(10) PRIMARY KEY,
    travel_date DATE,
    num_passenger NUMBER(3),
    destination VARCHAR2(100),
    metro_details VARCHAR2(255),
    reservation_status VARCHAR2(20),
    p_id NUMBER(10),
    FOREIGN KEY (p_id) REFERENCES passenger(p_id)
);
desc ticket;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **TICKET**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TICKET	TICKET_ID	Number	-	10	0	1	-	-	-
	TRAVEL_DATE	Date	7	-	-	-	✓	-	-
	NUM_PASSENGER	Number	-	3	0	-	✓	-	-
	DESTINATION	Varchar2	100	-	-	-	✓	-	-
	METRO_DETAILS	Varchar2	255	-	-	-	✓	-	-
	RESERVATION_STATUS	Varchar2	20	-	-	-	✓	-	-
	P_ID	Number	-	10	0	-	✓	-	-

1 - 7

Alter TABLE ticket

MODIFY (Travel_date

varchar2(20));

Payment table:

Autocommit Display 10

```
CREATE TABLE payment (
    payment_id NUMBER(10) PRIMARY KEY,
    online_payment VARCHAR2(20),
    total_payment NUMBER(10, 2),
    p_id NUMBER(10),
    FOREIGN KEY (p_id) REFERENCES passenger(p_id)
);
desc payment;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **PAYMENT**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PAYMENT	PAYMENT_ID	Number	-	10	0	1	-	-	-
	ONLINE_PAYMENT	Varchar2	20	-	-	-	✓	-	-
	TOTAL_PAYMENT	Number	-	10	2	-	✓	-	-
	P_ID	Number	-	10	0	-	✓	-	-

1 - 4

Station Table:

Autocommit Display 10

```
CREATE TABLE station (
    station_id NUMBER(5) PRIMARY KEY,
    station_name VARCHAR2(100),
    line_id NUMBER(5),
    route_id NUMBER(5),
    FOREIGN KEY (route_id) REFERENCES routes(route_id)
);
desc station;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **STATION**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
STATION	STATION_ID	Number	-	5	0	1	-	-	-
	STATION_NAME	Varchar2	100	-	-	-	✓	-	-
	LINE_ID	Number	-	5	0	-	✓	-	-
	ROUTE_ID	Number	-	5	0	-	✓	-	-

1 - 4

Zone Table:

```
CREATE TABLE zone (
    zone_id NUMBER(10) PRIMARY KEY,
    zone_name VARCHAR2(100),
    zone_code VARCHAR2(10),
    station_id NUMBER(10),
    FOREIGN KEY (station_id) REFERENCES station(station_id)
);
desc zone;
```

Results Explain Describe Saved SQL History

Object Type **TABLE** Object **ZONE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
ZONE	ZONE_ID	Number	-	10	0	1	-	-	-
	ZONE_NAME	Varchar2	100	-	-	-	✓	-	-
	ZONE_CODE	Varchar2	10	-	-	-	✓	-	-
	STATION_ID	Number	-	10	0	-	✓	-	-

1 - 4

Data Insearion:

Data for Metro_R

Autocommit Display 10

```
INSERT INTO metro_rail  VALUES (1, 'Metro A', 'Available', 'Engine');
INSERT INTO metro_rail  VALUES (2, 'Metro B', 'Reserved', 'Brake System');
INSERT INTO metro_rail  VALUES (3, 'Metro C', 'Available', 'Power Distribution');
INSERT INTO metro_rail  VALUES (4, 'Metro D', 'Reserved', 'HVAC System');
INSERT INTO metro_rail  VALUES (5, 'Metro E', 'Available', 'Lighting System');
INSERT INTO metro_rail  VALUES (6, 'Metro F', 'Available', 'Communication System');
INSERT INTO metro_rail  VALUES (7, 'Metro G', 'Reserved', 'Wheel Assembly');
INSERT INTO metro_rail  VALUES (8, 'Metro H', 'Available', 'Door System');
INSERT INTO metro_rail  VALUES (9, 'Metro I', 'Available', 'Safety System');
INSERT INTO metro_rail  VALUES (10, 'Metro J', 'Reserved', 'Control System');

select * from metro_rail;
```

Results Explain Describe Saved SQL History

METRO_ID	METRO_NAME	RESERVATION	COMPONENT
1	Metro A	Available	Engine
2	Metro B	Reserved	Brake System
3	Metro C	Available	Power Distribution
5	Metro E	Available	Lighting System
6	Metro F	Available	Communication System
7	Metro G	Reserved	Wheel Assembly
8	Metro H	Available	Door System
9	Metro I	Available	Safety System
10	Metro J	Reserved	Control System
4	Metro D	Reserved	HVAC System

10 rows returned in 0.00 seconds CSV Export

Data for Routes table :

Autocommit Display 10

```
INSERT INTO routes VALUES (1, 101, 1);
INSERT INTO routes VALUES (2, 102, 2);
INSERT INTO routes VALUES (3, 103, 3);
INSERT INTO routes VALUES (4, 104, 4);
INSERT INTO routes VALUES (5, 105, 5);
INSERT INTO routes VALUES (6, 106, 6);
INSERT INTO routes VALUES (7, 107, 7);
INSERT INTO routes VALUES (8, 108, 8);
INSERT INTO routes VALUES (9, 109, 9);
INSERT INTO routes VALUES (10, 110, 10);

select * from routes;
```

Results Explain Describe Saved SQL History

ROUTE_ID	DESTINATION_ID	METRO_ID
1	101	1
2	102	2
3	103	3
5	105	5
6	106	6
7	107	7
8	108	8
9	109	9
10	110	10
4	104	4

10 rows returned in 0.02 seconds CSV Export

Data for Passenger table :

Autocommit Display 10

```
INSERT INTO passenger VALUES (1, 'Alice', 25, 'F', '123-456-7890', 1);
INSERT INTO passenger VALUES (2, 'Bob', 30, 'M', '987-654-3210', 2);
INSERT INTO passenger VALUES (3, 'Charlie', 40, 'M', '111-222-3333', 3);
INSERT INTO passenger VALUES (4, 'David', 22, 'M', '444-555-6666', 4);
INSERT INTO passenger VALUES (5, 'Eve', 28, 'F', '777-888-9999', 5);
INSERT INTO passenger VALUES (6, 'Frank', 35, 'M', '222-333-4444', 6);
INSERT INTO passenger VALUES (7, 'Grace', 50, 'F', '555-666-7777', 7);
INSERT INTO passenger VALUES (8, 'Hannah', 27, 'F', '888-999-0000', 8);
INSERT INTO passenger VALUES (9, 'Ian', 45, 'M', '333-444-5555', 9);
INSERT INTO passenger VALUES (10, 'Jessica', 32, 'F', '999-000-1111', 10);

select * from passenger;
```

Results Explain Describe Saved SQL History

P_ID	P_NAME	P_AGE	GENDER	P_PHONE	METRO_ID
1	Alice	25	F	123-456-7890	1
2	Bob	30	M	987-654-3210	2
3	Charlie	40	M	111-222-3333	3
5	Eve	28	F	777-888-9999	5
6	Frank	35	M	222-333-4444	6
7	Grace	50	F	555-666-7777	7
8	Hannah	27	F	888-999-0000	8
9	Ian	45	M	333-444-5555	9
10	Jessica	32	F	999-000-1111	10
4	David	22	M	444-555-6666	4

10 rows returned in 0.00 seconds CSV Export

Data for Ticket table :

Autocommit Display 10

```
INSERT INTO ticket VALUES (1, '2024-05-17', 1, 'Station A', 'Metro A, Carriage 1, Seat 5', 'Confirmed', 1);
INSERT INTO ticket VALUES (2, '2024-05-17', 2, 'Station B', 'Metro B, Carriage 2, Seat 10', 'Confirmed', 2);
INSERT INTO ticket VALUES (3, '2024-05-18', 1, 'Station C', 'Metro C, Carriage 3, Seat 15', 'Confirmed', 3);
INSERT INTO ticket VALUES (4, '2024-05-18', 1, 'Station D', 'Metro D, Carriage 4, Seat 20', 'Confirmed', 4);
INSERT INTO ticket VALUES (5, '2024-05-19', 1, 'Station E', 'Metro E, Carriage 5, Seat 25', 'Confirmed', 5);
INSERT INTO ticket VALUES (6, '2024-05-19', 3, 'Station F', 'Metro F, Carriage 6, Seats 30-32', 'Confirmed', 6);
INSERT INTO ticket VALUES (7, '2024-05-20', 2, 'Station G', 'Metro G, Carriage 7, Seats 35-36', 'Confirmed', 7);
INSERT INTO ticket VALUES (8, '2024-05-20', 1, 'Station H', 'Metro H, Carriage 8, Seat 40', 'Confirmed', 8);
INSERT INTO ticket VALUES (9, '2024-05-21', 1, 'Station I', 'Metro I, Carriage 9, Seat 45', 'Confirmed', 9);
INSERT INTO ticket VALUES (10, '2024-05-21', 2, 'Station J', 'Metro J, Carriage 10, Seats 50-51', 'Confirmed', 10);

select * from ticket;
```

Results Explain Describe Saved SQL History

TICKET_ID	TRAVEL_DATE	NUM_PASSENGER	DESTINATION	METRO_DETAILS	RESERVATION_STATUS	P_ID
1	2024-05-17	1	Station A	Metro A, Carriage 1, Seat 5	Confirmed	1
2	2024-05-17	2	Station B	Metro B, Carriage 2, Seat 10	Confirmed	2
3	2024-05-18	1	Station C	Metro C, Carriage 3, Seat 15	Confirmed	3
4	2024-05-18	1	Station D	Metro D, Carriage 4, Seat 20	Confirmed	4
5	2024-05-19	1	Station E	Metro E, Carriage 5, Seat 25	Confirmed	5
6	2024-05-19	3	Station F	Metro F, Carriage 6, Seats 30-32	Confirmed	6
7	2024-05-20	2	Station G	Metro G, Carriage 7, Seats 35-36	Confirmed	7
8	2024-05-20	1	Station H	Metro H, Carriage 8, Seat 40	Confirmed	8
9	2024-05-21	1	Station I	Metro I, Carriage 9, Seat 45	Confirmed	9
10	2024-05-21	2	Station J	Metro J, Carriage 10, Seats 50-51	Confirmed	10

10 rows returned in 0.00 seconds CSV Export

Data for Payment table :

Autocommit Display 10

```
INSERT INTO payment VALUES (1, 'Credit Card', 50.00, 1);
INSERT INTO payment VALUES (2, 'Debit Card', 30.00, 2);
INSERT INTO payment VALUES (3, 'Credit Card', 25.00, 3);
INSERT INTO payment VALUES (4, 'Debit Card', 40.00, 4);
INSERT INTO payment VALUES (5, 'Credit Card', 20.00, 5);
INSERT INTO payment VALUES (6, 'Debit Card', 35.00, 6);
INSERT INTO payment VALUES (7, 'Credit Card', 15.00, 7);
INSERT INTO payment VALUES (8, 'Debit Card', 60.00, 8);
INSERT INTO payment VALUES (9, 'Credit Card', 55.00, 9);
INSERT INTO payment VALUES (10, 'Debit Card', 70.00, 10);

select * from payment;
```

Results Explain Describe Saved SQL History

PAYMENT_ID	ONLINE_PAYMENT	TOTAL_PAYMENT	P_ID
1	Credit Card	50	1
2	Debit Card	30	2
3	Credit Card	25	3
4	Debit Card	40	4
5	Credit Card	20	5
6	Debit Card	35	6
7	Credit Card	15	7
8	Debit Card	60	8
9	Credit Card	55	9
10	Debit Card	70	10

10 rows returned in 0.00 seconds CSV Export

Data for Station table :

Autocommit Display 10

```
INSERT INTO station VALUES (1, 'Station A', 101, 1);
INSERT INTO station VALUES (2, 'Station B', 102, 2);
INSERT INTO station VALUES (3, 'Station C', 103, 3);
INSERT INTO station VALUES (4, 'Station D', 104, 4);
INSERT INTO station VALUES (5, 'Station E', 105, 5);
INSERT INTO station VALUES (6, 'Station F', 106, 6);
INSERT INTO station VALUES (7, 'Station G', 107, 7);
INSERT INTO station VALUES (8, 'Station H', 108, 8);
INSERT INTO station VALUES (9, 'Station I', 109, 9);
INSERT INTO station VALUES (10, 'Station J', 110, 10);

select * from station;
```

Results Explain Describe Saved SQL History

STATION_ID	STATION_NAME	LINE_ID	ROUTE_ID
1	Station A	101	1
2	Station B	102	2
3	Station C	103	3
4	Station D	104	4
5	Station E	105	5
6	Station F	106	6
7	Station G	107	7
8	Station H	108	8
9	Station I	109	9
10	Station J	110	10

Data for Zone table :

Autocommit Display 10

```
INSERT INTO zone VALUES (1, 'Zone A', 'ZA001', 1);
INSERT INTO zone VALUES (2, 'Zone B', 'ZA002', 2);
INSERT INTO zone VALUES (3, 'Zone C', 'ZA003', 3);
INSERT INTO zone VALUES (4, 'Zone D', 'ZA004', 4);
INSERT INTO zone VALUES (5, 'Zone E', 'ZA005', 5);
INSERT INTO zone VALUES (6, 'Zone F', 'ZA006', 6);
INSERT INTO zone VALUES (7, 'Zone G', 'ZA007', 7);
INSERT INTO zone VALUES (8, 'Zone H', 'ZA008', 8);
INSERT INTO zone VALUES (9, 'Zone I', 'ZA009', 9);
INSERT INTO zone VALUES (10, 'Zone J', 'ZA010', 10);

select * from zone;
```

Results Explain Describe Saved SQL History

ZONE_ID	ZONE_NAME	ZONE_CODE	STATION_ID
1	Zone A	ZA001	1
2	Zone B	ZA002	2
3	Zone C	ZA003	3
4	Zone D	ZA004	4
5	Zone E	ZA005	5
6	Zone F	ZA006	6
7	Zone G	ZA007	7
8	Zone H	ZA008	8
9	Zone I	ZA009	9
10	Zone J	ZA010	10

10 rows returned in 0.00 seconds CSV Export

Query Writing:

Single-Row Functions:

1. **Show the available reservation of metro rail.**

```
select*from metro_rail where reservation ='Available';
```

Results Explain Describe Saved SQL History

METRO_ID	METRO_NAME	RESERVATION	COMPONENT
1	Metro A	Available	Engine
3	Metro C	Available	Power Distribution
5	Metro E	Available	Lighting System
6	Metro F	Available	Communication System
8	Metro H	Available	Door System
9	Metro I	Available	Safety System

6 rows returned in 0.02 seconds CSV Export

Language: en-us

2.Show passenger those who are more than 30 years.

Autocommit Display 10

```
select*from passenger where p_age>30;
```

Results Explain Describe Saved SQL History

P_ID	P_NAME	P_AGE	GENDER	P_PHONE	METRO_ID
3	Charlie	40	M	111-222-3333	3
6	Frank	35	M	222-333-4444	6
7	Grace	50	F	555-666-7777	7
9	Ian	45	M	333-444-5555	9
10	Jessica	32	F	999-000-1111	10

5 rows returned in 0.01 seconds CSV Export

group function:

1.Show the total payment.

Autocommit Display 10

```
select  sum(total_payment) from payment;
```

Results Explain Describe Saved SQL History

SUM(TOTAL_PAYMENT)
400

1 rows returned in 0.01 seconds CSV Export

2. Show average of total payment.

Autocommit Display 10

```
select avg(total_payment) as "average payment"
from payment;
```

Results Explain Describe Saved SQL History

Average Payment
40

1 rows returned in 0.00 seconds CSV Export

Single row subquery:

1.Show the female passenger.

Autocommit Display 10

```
select * from passenger where gender=(select gender from passenger where p_id=1);
```

Results Explain Describe Saved SQL History

P_ID	P_NAME	P_AGE	GENDER	P_PHONE	METRO_ID
1	Alice	25	F	123-456-7890	1
5	Eve	28	F	777-888-9999	5
7	Grace	50	F	555-666-7777	7
8	Hannah	27	F	888-999-0000	8
10	Jessica	32	F	999-000-1111	10

5 rows returned in 0.02 seconds CSV Export

2.Show the destination details for num_passenger 3.

Autocommit Display 10

```
Select *
from ticket
where destination = (select destination
                     from ticket
                     where num_passenger = 3);
```

Results Explain Describe Saved SQL History

TICKET_ID	TRAVEL_DATE	NUM_PASSENGER	DESTINATION	METRO_DETAILS	RESERVATION_STATUS	P_ID
6	2024-05-19	3	Station F	Metro F, Carriage 6, Seats 30-32	Confirmed	6

1 rows returned in 0.00 seconds CSV Export

Multiple row subquery:

1. Show the passenger with an age less than any male passenger's age while passenger's age while excluding male passengers from the result set.W

Autocommit Display 10

```
select p_id, p_name, gender, p_age, p_phone
from passenger
where p_age  < ANY (select p_age
                    from passenger
                    where gender = 'M')
                    And gender <> 'M';
```

Results Explain Describe Saved SQL History

P_ID	P_NAME	GENDER	P_AGE	P_PHONE
1	Alice	F	25	123-456-7890
5	Eve	F	28	777-888-9999
8	Hannah	F	27	888-999-0000
10	Jessica	F	32	999-000-1111

4 rows returned in 0.02 seconds CSV Export

Joining

1. What specific information are you trying to retrieve by joining the 'metro_id' and passenger tables on the metro_id and p_id column?

```
select*from metro_rail m,passenger p
where m.metro_id = p.p_id;
```

Results Explain Describe Saved SQL History

METRO_ID	METRO_NAME	RESERVATION	COMPONENT	P_ID	P_NAME	P_AGE	GENDER	P_PHONE	METRO_ID
1	Metro A	Available	Engine	1	Alice	25	F	123-456-7890	1
2	Metro B	Reserved	Brake System	2	Bob	30	M	987-654-3210	2
3	Metro C	Available	Power Distribution	3	Charlie	40	M	111-222-3333	3
5	Metro E	Available	Lighting System	5	Eve	28	F	777-888-9999	5
6	Metro F	Available	Communication System	6	Frank	35	M	222-333-4444	6
7	Metro G	Reserved	Wheel Assembly	7	Grace	50	F	555-666-7777	7
8	Metro H	Available	Door System	8	Hannah	27	F	888-999-0000	8
9	Metro I	Available	Safety System	9	Ian	45	M	333-444-5555	9
10	Metro J	Reserved	Control System	10	Jessica	32	F	999-000-1111	10
4	Metro D	Reserved	HVAC System	4	David	22	M	444-555-6666	4

10 rows returned in 0.00 seconds CSV Export

2. What insights are you hoping to gain by performing a full outer join between the passenger and payment tables on the p_id column?

```
select*
from passenger full outer join payment
on passenger.p_id = payment.p_id;
```

Results Explain Describe Saved SQL History

P_ID	P_NAME	P_AGE	GENDER	P_PHONE	METRO_ID	PAYMENT_ID	ONLINE_PAYMENT	TOTAL_PAYMENT	P_ID
1	Alice	25	F	123-456-7890	1	1	Credit Card	50	1
2	Bob	30	M	987-654-3210	2	2	Debit Card	30	2
3	Charlie	40	M	111-222-3333	3	3	Credit Card	25	3
4	David	22	M	444-555-6666	4	4	Debit Card	40	4
5	Eve	28	F	777-888-9999	5	5	Credit Card	20	5
6	Frank	35	M	222-333-4444	6	6	Debit Card	35	6
7	Grace	50	F	555-666-7777	7	7	Credit Card	15	7
8	Hannah	27	F	888-999-0000	8	8	Debit Card	60	8
9	Ian	45	M	333-444-5555	9	9	Credit Card	55	9
10	Jessica	32	F	999-000-1111	10	10	Debit Card	70	10

10 rows returned in 0.01 seconds CSV Export

3. Retrieve all records from the ticket table and matching records from the passenger table based on the P_id column, where available?

```
select*
from passenger right outer join ticket
on passenger.p_id = ticket.p_id;
```

Results Explain Describe Saved SQL History

P_ID	P_NAME	P_AGE	GENDER	P_PHONE	METRO_ID	TICKET_ID	TRAVEL_DATE	NUM_PASSENGER	DESTINATION	METRO_DETAILS	RESERVATION_STATUS	P_ID
1	Alice	25	F	123-456-7890	1	1	2024-05-17	1	Station A	Metro A, Carriage 1, Seat 5	Confirmed	1
2	Bob	30	M	987-654-3210	2	2	2024-05-17	2	Station B	Metro B, Carriage 2, Seat 10	Confirmed	2
3	Charlie	40	M	111-222-3333	3	3	2024-05-18	1	Station C	Metro C, Carriage 3, Seat 15	Confirmed	3
5	Eve	28	F	777-888-9999	5	5	2024-05-19	1	Station E	Metro E, Carriage 5, Seat 25	Confirmed	5
6	Frank	35	M	222-333-4444	6	6	2024-05-19	3	Station F	Metro F, Carriage 6, Seats 30-32	Confirmed	6
7	Grace	50	F	555-666-7777	7	7	2024-05-20	2	Station G	Metro G, Carriage 7, Seats 35-36	Confirmed	7
8	Hannah	27	F	888-999-0000	8	8	2024-05-20	1	Station H	Metro H, Carriage 8, Seat 40	Confirmed	8
9	Ian	45	M	333-444-5555	9	9	2024-05-21	1	Station I	Metro I, Carriage 9, Seat 45	Confirmed	9
10	Jessica	32	F	999-000-1111	10	10	2024-05-21	2	Station J	Metro J, Carriage 10, Seats 50-51	Confirmed	10
4	David	22	M	444-555-6666	4	4	2024-05-18	1	Station D	Metro D, Carriage 4, Seat 20	Confirmed	4

10 rows returned in 0.02 seconds CSV Export

View :

1. Serve the context in the context of the database schema and querying operations.

Autocommit Display 10

```
create View passengerView
AS SELECT p_id AS passenger_id, p_name AS passenger_name from passenger;
```

Results Explain Describe Saved SQL History

```
View created.
```

```
desc passengerview;
```

Results Explain Describe Saved SQL History

Object Type **VIEW** Object **PASSENGERVIEW**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PASSENGERVIEW	PASSENGER_ID	Number	-	10	0	-	-	-	-
	PASSENGER_NAME	Varchar2	100	-	-	-	✓	-	-

1 - 2

Autocommit Display 10

```
select*from passengerview;
```

Results Explain Describe Saved SQL History

PASSENGER_ID	PASSENGER_NAME
1	Alice
2	Bob
3	Charlie
5	Eve
6	Frank
7	Grace
8	Hannah
9	Ian
10	Jessica
4	David

2.

create VIEW ticketView

AS select ticket_id "Ticket_number" , travel_date "date", destination , metro_details "metro_name" FROM Ticket where num_passengers = 1;

Results Explain Describe Saved SQL History

View created.

0.09 seconds

desc ticketview;

Results Explain Describe Saved SQL History

Object Type **VIEW** Object **TICKETVIEW**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TICKETVIEW	Ticket_number	Number	-	10	0	-	-	-	-
	date	Date	7	-	-	-	✓	-	-
	DESTINATION	Varchar2	100	-	-	-	✓	-	-
	metro_name	Varchar2	100	-	-	-	✓	-	-

1 - 4

select* from ticketview;

Results Explain Describe Saved SQL History

Ticket_number	Date	DESTINATION	Metro_name
1	15-MAY-24	End1	Metro1
3	17-MAY-24	End3	Metro3
4	18-MAY-24	End4	Metro4
6	20-MAY-24	End6	Metro6
8	22-MAY-24	End8	Metro8
9	23-MAY-24	End9	Metro9

6 rows returned in 0.02 seconds CSV Export

relation algebra:

Question 1. Show how many metro rail is there?

π Metro_Id (MetroRail)

Question 2 : How many passenger are there?

π P_Id (Passenger)

Question 3: Show tickets where the number of passengers is greater than 2.

σ(No_Of_Passenger > 2)(Ticket)

Queation 4: Project only the Ticket ID.

π(Ticket_Id)(Ticket)

Question 5: Select passengers with a phone number.

σ(P_Phone IS NOT NULL)(Passenger)

Conclusion:

The Metro Rail Database Project has significantly transformed the management and operation of metro rail networks, establishing a centralized repository for crucial data storage and access. Its user-friendly interface has greatly improved passenger experience by providing real-time updates, easy access to schedules, routes, and fares. Moreover, administrative tools empower system administrators to efficiently manage and update the database, ensuring accuracy and timeliness in information dissemination. Looking forward, continuous improvement is essential, incorporating new technologies and meeting evolving passenger needs. Expansion to accommodate the growth of the metro network, integration with other transportation systems, and sustainability initiatives are crucial steps. Additionally, collaboration with stakeholders and investing in research and development for innovative solutions will further optimize system performance and enhance passenger satisfaction.